ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)

ACADEX PRACTICE EXAMINATION

A-LEVEL · FURTHER MATHEMATICS

MATHEMATICS 9187/1

PAPER 1 November 2023

3 hours

PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

Additional materials: A calculator may be used unless a question forbids it.
INSTRUCTIONS: Answer all questions. Full marks are not given for answers only — show method.

1. $z = 3 + 4i$. Find $|z|$, arg z, and $\bar{z}$. Express $1/z$ in the form $a+bi$. **[10]**

2. $A = (2\ 1\ ;\ 0\ 3)$. Find det A, A^{-1}, and A^2. **[10]**

3. Prove by induction that $1 + 2 + \ldots + n = n(n+1)/2$ for $n \geq 1$. **[10]**

4. Write $1 + i$ in modulus-argument form and find $(1+i)^4$ using De Moivre. **[8]**

5. Use identities to simplify $\sin 3\theta$ in terms of $\sin \theta$ (or state $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$). **[8]**

6. Show that $(i+j)\times(i+2j+k) = i - j + k$ or compute the cross product of $a=(1,1,0)$, $b=(1,2,1)$. **[8]**

7. Standard Further Pure question on Maclaurin series. Show full working. **[8]**

8. Standard Further Pure question on Reduction formula sketch. Show full working. **[8]**

9. Standard Further Pure question on Eigenvalues of (2,1;1,2). Show full working. **[8]**

10. Standard Further Pure question on Second order DE $y''+y=0$. Show full working. **[8]**

11. Standard Further Pure question on Further partial fractions. Show full working. **[8]**

12. Standard Further Pure question on Argand diagram loci. Show full working. **[8]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).